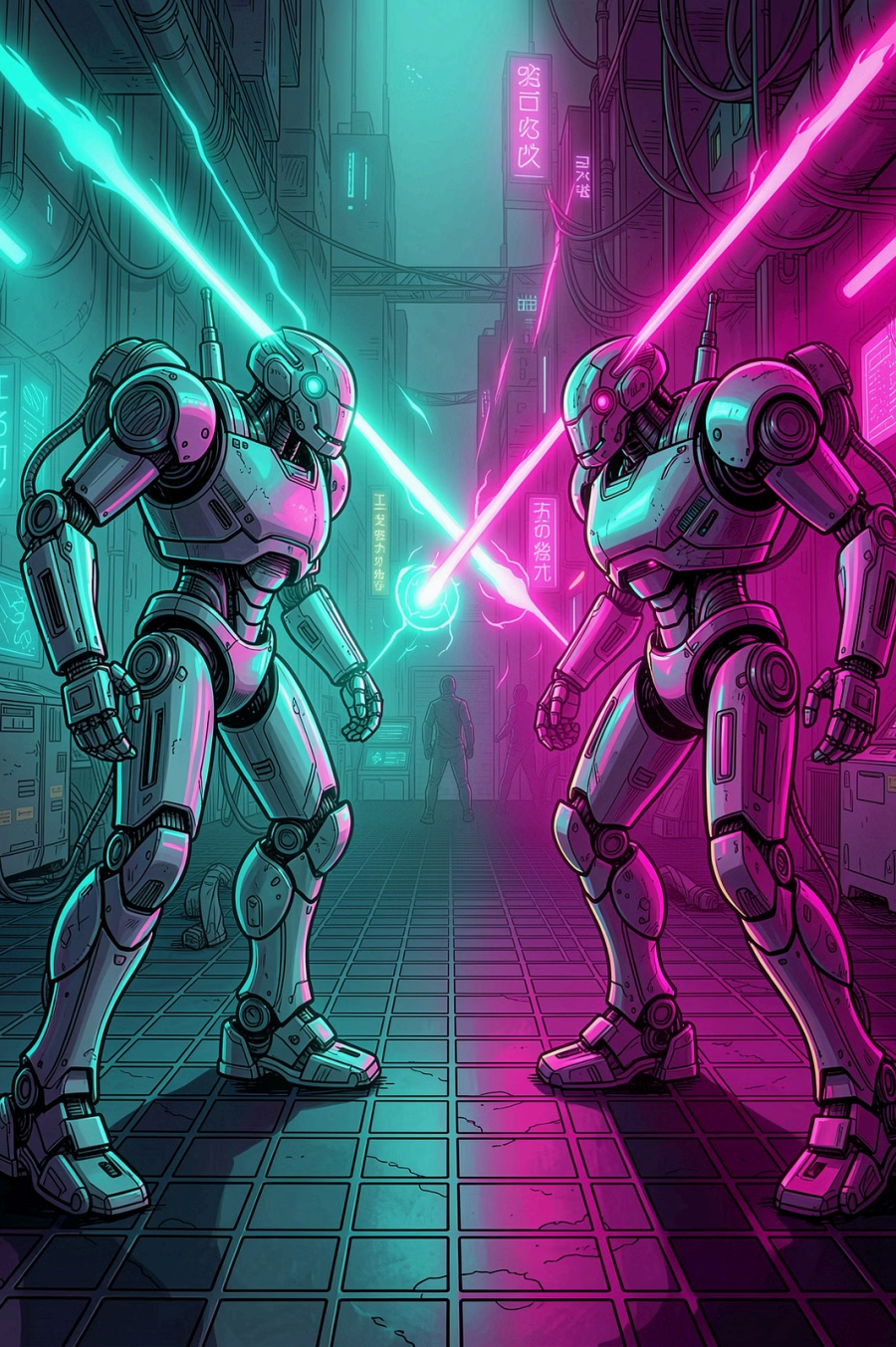




BOOLEAN BATTLE ARENA

A Real-Time Autonomous Combat Simulation Using Boolean Logic

REACT

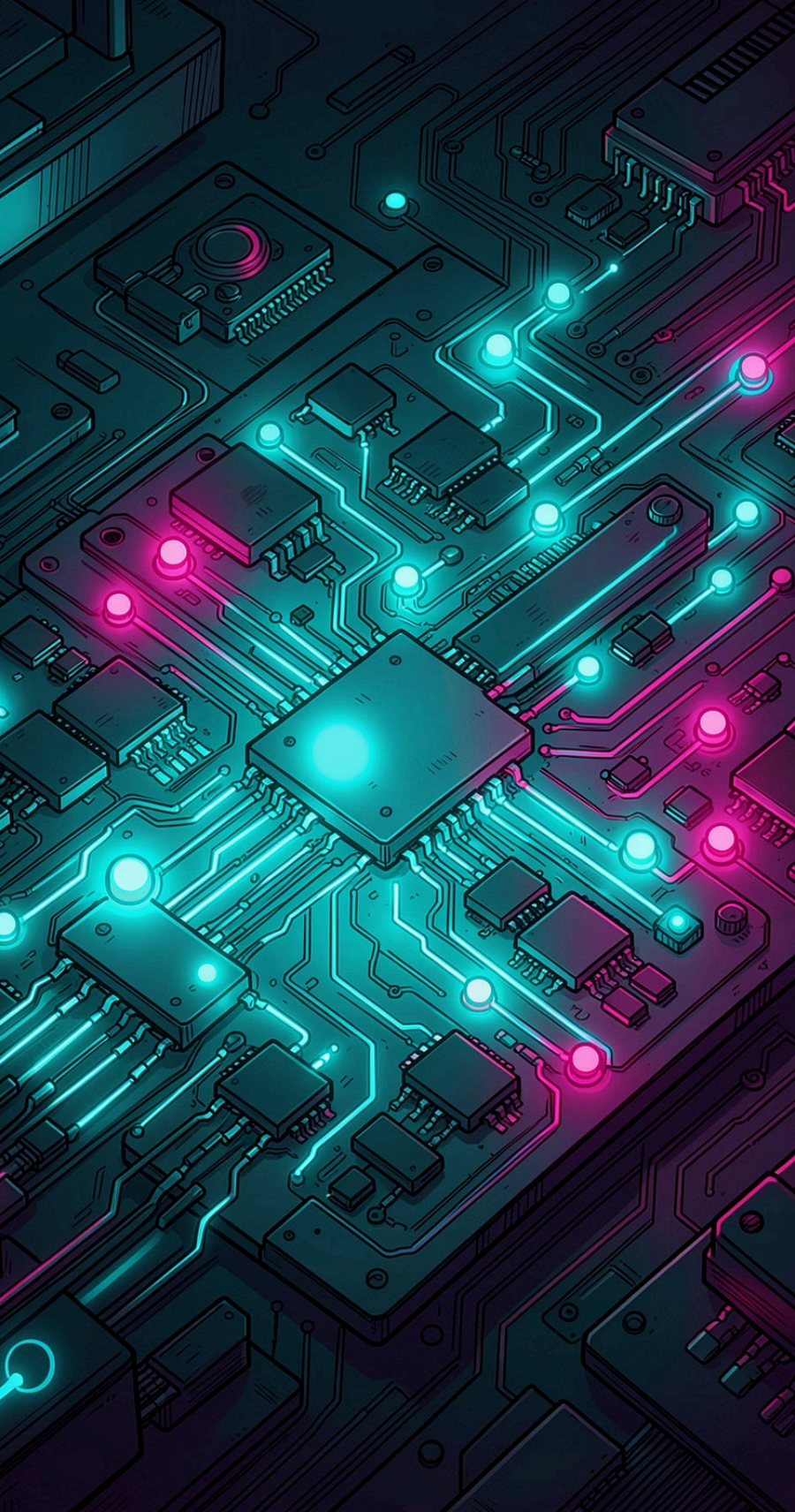
CANVAS

BOOLEAN LOGIC

PHYSICS ENGINE

SIMULATION

[Author Name] · [Company / Internship] · [Date]



Project Objective

Create autonomous bots that make real-time combat decisions using **pure Boolean logic** — no machine learning, no neural networks.

Simulate Combat

Emergent behaviors from deterministic rule-based systems

Sensor Systems

Bots perceive their environment through discrete Boolean outputs

Visualize Behavior

Real-time Canvas rendering at ~60 FPS

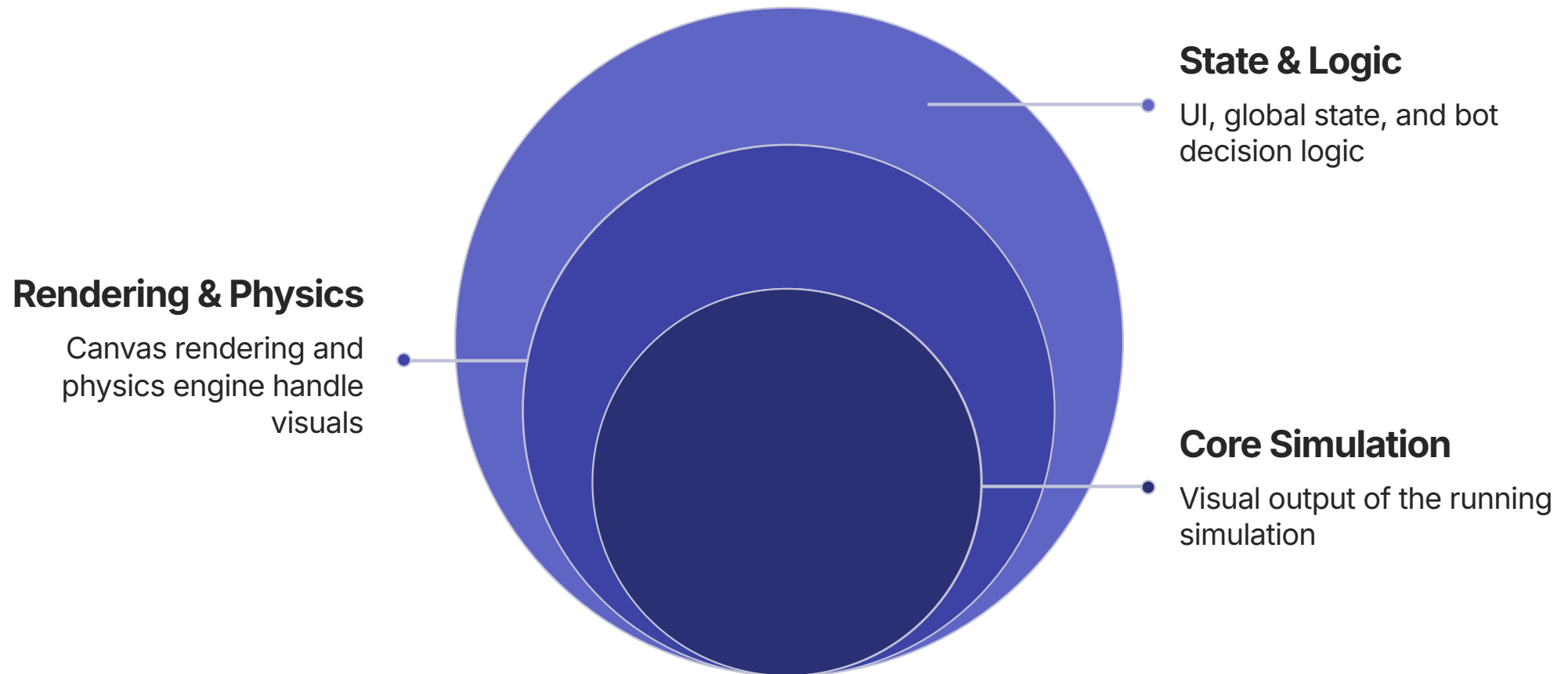
Optimize Rendering

Efficient pipeline using React + Canvas separation

How simple **TRUE / FALSE** conditions create complex, emergent combat intelligence

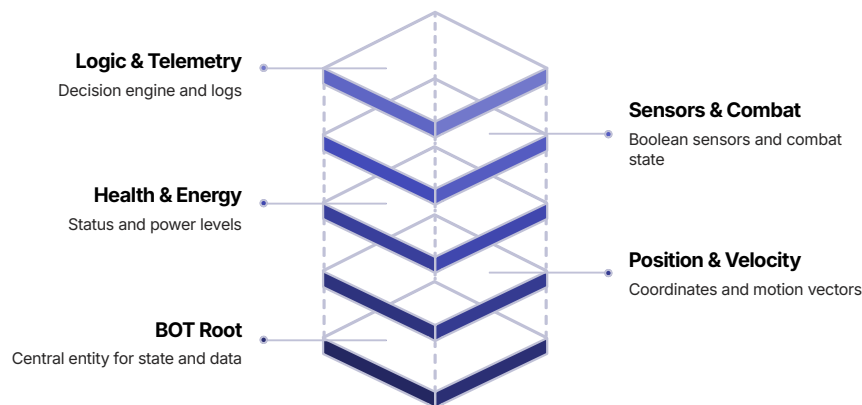
High-Level Architecture

A fully modular, layered architecture designed for scalability and maintainability — each layer has a single, clearly defined responsibility.



Each layer is independently testable. The Canvas renderer bypasses React's virtual DOM for performance-critical frame updates, while global state is managed via a single reducer for predictability.

Bot Architecture & Sensor System



Boolean Sensor Outputs

Each sensor returns a discrete TRUE/FALSE signal that feeds directly into the decision engine:

i Boolean outputs become direct inputs for the decision engine — no intermediate translation required.

enemySeen = TRUE

An enemy bot is within sensor cone range

nearWall = FALSE

No boundary collision imminent

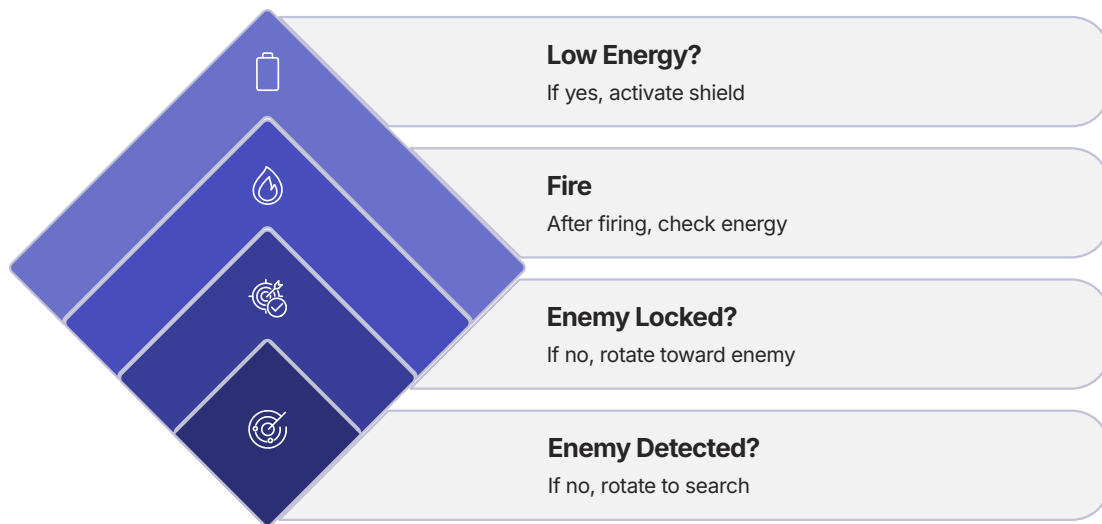
lowEnergy = TRUE

Energy below critical threshold — shield unavailable

enemyLocked = FALSE

Target not yet in firing arc

Boolean Decision Engine



Rule-Based Intelligence

Pure conditional logic chains produce surprisingly sophisticated combat behavior — without a single trained model.

"Simple Boolean rules create emergent combat behavior indistinguishable from intentional strategy."



No ML

Zero training data required



No Neural Nets

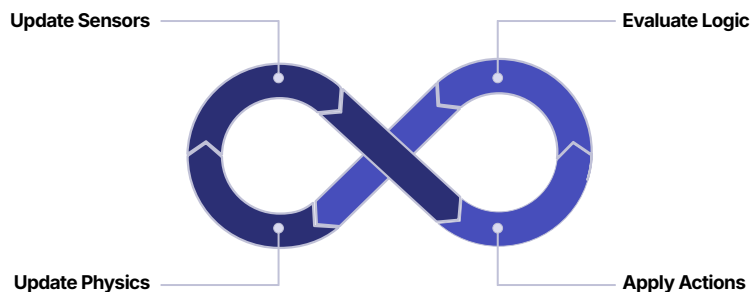
Fully deterministic and debuggable



Pure Logic

Emergent behavior from Boolean rules

Real-Time Simulation Pipeline



Performance Optimizations

The simulation sustains ~60 FPS by aggressively separating rendering from state logic:

→ **requestAnimationFrame**

Native browser scheduling — no `setInterval` drift

→ **useRef for Mutable State**

Prevents unnecessary re-renders on each physics tick

→ **Canvas over React DOM**

Bypasses virtual DOM diffing for frame-critical drawing

→ **Telemetry Throttling**

UI dashboards update at reduced frequency — decoupled from render loop

Challenges, Results & Future Scope

Challenges & Solutions

Wall Collision Loops

Resolved with escape cooldown logic — bots can't re-trigger evasion within a fixed window

Excessive Re-renders

Decoupled Canvas render loop from React component tree using `useRef`

Physics & Combat Tuning

Iterative balancing of speed, energy, and damage parameters for fair emergent gameplay

Key Takeaway: Complex autonomous behavior can emerge from simple Boolean decision systems.

Future Improvements

01

Reinforcement Learning

Replace static rules with learned policies

02

Evolutionary Algorithms

Bots that improve across simulation generations

03

Team Battles & Adaptive AI

Multi-bot coordination and dynamic difficulty

04

Analytics Dashboards

Real-time telemetry and match replay systems

